



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Case Study No.: 10

1. Case Study Title: A college maintains student registration data for various campus events in a JSON file. The event coordinator wants a web page that can load this information dynamically and display it in a table without manually entering the data. The student JSON file consists of Student Name, PRN, Department, Year, Event Name, and Registration Status (in a drop-down we can filter/search based on registered, pending, or cancelled) using `fetch()` and `$.getJSON()`, display an appropriate message if JSON data cannot be loaded and no student matches the search/filter criteria.

2. Case Study Program Code:

```
/* Sonal Nakhate 24070521199 */
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Event Registration Table</title>

  <!-- Include jQuery library for $.getJSON() capability -->
  <script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>

<style>
  body {
    font-family: Arial, sans-serif;
    margin: 20px;
    background-color: #f8f9fa;
    color: #333;
  }
```



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```
h2 {
  color: #222;
  margin-bottom: 5px;
}

p.description {
  color: #666;
  font-size: 14px;
  margin-top: 0;
  margin-bottom: 20px;
}

/* Controls Bar Styling */
.controls {
  background-color: #ffffff;
  padding: 15px;
  border: 1px solid #ccc;
  border-radius: 6px;
  margin-bottom: 15px;
  display: flex;
  gap: 15px;
  align-items: center;
  flex-wrap: wrap;
}

.control-group {
  display: flex;
  align-items: center;
  gap: 8px;
}

label {
  font-weight: bold;
  font-size: 14px;
```



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```
}

input[type="text"], select {
  padding: 6px 10px;
  border: 1px solid #ccc;
  border-radius: 4px;
  font-size: 14px;
}

button {
  padding: 7px 12px;
  background-color: #007bff;
  color: white;
  border: none;
  border-radius: 4px;
  cursor: pointer;
  font-size: 14px;
}

button:hover {
  background-color: #0056b3;
}

button.jquery-btn {
  background-color: #28a745;
}

button.jquery-btn:hover {
  background-color: #218838;
}

/* Error Alert Box */
.error-msg {
  background-color: #f8d7da;
```



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```
color: #721c24;
padding: 12px;
border: 1px solid #f5c6cb;
border-radius: 4px;
margin-bottom: 15px;
display: none;
font-size: 14px;
}

/* Table Styling */
table {
  width: 100%;
  border-collapse: collapse;
  background-color: #ffffff;
  border: 1px solid #ccc;
}

th, td {
  padding: 10px;
  text-align: left;
  border: 1px solid #ddd;
  font-size: 14px;
}

th {
  background-color: #e9ecef;
}

tr:nth-child(even) {
  background-color: #f9f9f9;
}

/* Plain Text Status Badge Styling */
.status-badge {
```



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```
display: inline-block;
padding: 4px 8px;
font-size: 12px;
font-weight: bold;
border-radius: 4px;
}

.status-registered {
background-color: #d4edda;
color: #155724;
border: 1px solid #c3e6cb;
}

.status-pending {
background-color: #fff3cd;
color: #856404;
border: 1px solid #ffeeba;
}

.status-cancelled {
background-color: #f8d7da;
color: #721c24;
border: 1px solid #f5c6cb;
}

.no-results-td {
text-align: center;
color: #856404;
background-color: #fff3cd;
padding: 15px;
font-weight: bold;
}
</style>
</head>
```



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<body>

<h2>Event Registration Table</h2>

<p class="description">Student event registration manager with real-time text search and status filter dropdown.</p>

<!-- Error message display area (hidden by default) -->

<div id="errorMessage" class="error-msg"></div>

<!-- Filter & Fetch Controls -->

<div class="controls">

<div class="control-group">

<label for="searchInput">Search:</label>

<input type="text" id="searchInput" placeholder="Search name, PRN, event...">

</div>

<div class="control-group">

<label for="statusFilter">Status Filter:</label>

<select id="statusFilter">

<option value="All">All Statuses</option>

<option value="Registered">Registered</option>

<option value="Pending">Pending</option>

<option value="Cancelled">Cancelled</option>

</select>

</div>

<button onclick="loadWithFetch()">Try Load via fetch()</button>

<button class="jquery-btn" onclick="loadWithJQuery()">Try Load via \$.getJSON()</button>

</div>

<!-- Registration Table -->

<table>



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```
<thead>
  <tr>
    <th>Student Name</th>
    <th>PRN</th>
    <th>Department</th>
    <th>Year</th>
    <th>Event Name</th>
    <th>Registration Status</th>
  </tr>
</thead>
<tbody id="studentTableBody">
  <!-- Dynamically filled by JavaScript -->
</tbody>
</table>

<script>
  // Default master dataset loaded directly on startup
  const defaultStudents = [
    {
      "studentName": "Aarav Sharma",
      "PRN": "2024PRN001",
      "department": "Computer Science",
      "year": "3rd Year",
      "eventName": "AI Tech Summit",
      "registrationStatus": "Registered"
    },
    {
      "studentName": "Priya Patel",
      "PRN": "2024PRN002",
      "department": "Information Technology",
      "year": "2nd Year",
      "eventName": "Cyber Security Workshop",
      "registrationStatus": "Pending"
    },
  ],
```



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```
{
  "studentName": "Rohan Deshmukh",
  "PRN": "2024PRN003",
  "department": "Electronics",
  "year": "4th Year",
  "eventName": "Robotics League 2026",
  "registrationStatus": "Cancelled"
},
{
  "studentName": "Ananya Roy",
  "PRN": "2024PRN004",
  "department": "Computer Science",
  "year": "1st Year",
  "eventName": "CodeSprint Hackathon",
  "registrationStatus": "Registered"
},
{
  "studentName": "Vikram Singh",
  "PRN": "2024PRN005",
  "department": "Mechanical Eng.",
  "year": "3rd Year",
  "eventName": "CAD Expo",
  "registrationStatus": "Pending"
}
];

// Current student list displayed in memory
let students = [...defaultStudents];

// Method 1: Load external file with standard fetch() API
function loadWithFetch() {
  hideError();
  fetch('students.json')
    .then(response => {
```



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```
        if (!response.ok) {
            throw new Error(`HTTP ${response.status}: Failed to locate
'students.json' file.`);
        }
        return response.json();
    })
    .then(data => {
        students = data;
        applyFilterAndSearch();
    })
    .catch(error => {
        showError("Error loading 'students.json' via fetch(): " +
error.message);
    });
}

// Method 2: Load external file with jQuery $.getJSON() API
function loadWithjQuery() {
    hideError();
    $.getJSON('students.json')
        .done(function(data) {
            students = data;
            applyFilterAndSearch();
        })
        .fail(function(jqxhr, textStatus, error) {
            showError("Error loading 'students.json' via $.getJSON(): File
not found or local browser restriction.");
        });
}

// Filter table by text search and dropdown status
function applyFilterAndSearch() {
    const searchText =
document.getElementById('searchInput').value.toLowerCase().trim();
```



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```
const selectedStatus =
document.getElementById('statusFilter').value;

const filteredStudents = students.filter(student => {
  const matchesSearch =
    student.studentName.toLowerCase().includes(searchText) ||
    student.PRN.toLowerCase().includes(searchText) ||
    student.department.toLowerCase().includes(searchText) ||
    student.eventName.toLowerCase().includes(searchText);

    const matchesStatus = (selectedStatus === "All") ||
(student.registrationStatus === selectedStatus);

  return matchesSearch && matchesStatus;
});

renderTable(filteredStudents);
}

// Render filtered student list into table rows
function renderTable(studentList) {
  const tbody = document.getElementById('studentTableBody');
  tbody.innerHTML = "";

  // Message when search or filter returns zero records
  if (studentList.length === 0) {
    tbody.innerHTML = `
      <tr>
        <td colspan="6" class="no-results-td">
          No student matches the search/filter criteria.
        </td>
      </tr>`;
    return;
  }
}
```



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```
studentList.forEach(student => {
    let statusClass = '';
    if (student.registrationStatus === 'Registered') statusClass =
'status-registered';
    else if (student.registrationStatus === 'Pending') statusClass =
'status-pending';
    else if (student.registrationStatus === 'Cancelled') statusClass =
'status-cancelled';

    const row = document.createElement('tr');
    row.innerHTML = `
        <td>${student.studentName}</td>
        <td>${student.PRN}</td>
        <td>${student.department}</td>
        <td>${student.year}</td>
        <td>${student.eventName}</td>
        <td>
            <span class="status-badge
${statusClass}">${student.registrationStatus}</span>
        </td>
    `;
    tbody.appendChild(row);
});

function showError(message) {
    const errBox = document.getElementById('errorMessage');
    errBox.textContent = message;
    errBox.style.display = 'block';
}

function hideError() {
    const errBox = document.getElementById('errorMessage');
```



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```
    errBox.style.display = 'none';
  }

  // Attach event listeners for search box and status dropdown
  document.getElementById('searchInput').addEventListener('input',
applyFilterAndSearch);
  document.getElementById('statusFilter').addEventListener('change',
applyFilterAndSearch);

  // Initial page startup: render table immediately from default array
  window.onload = function() {
    applyFilterAndSearch();
  };
</script>
</body>
</html>
```

3. Output:

/* Sonal Nakhate 24070521199 */

Event Registration Table

Student event registration manager with real-time text search and status filter dropdown.

Search:

Status Filter:

All Statuses

Try Load via fetch()

Try Load via \$.getJSON()

Student Name	PRN	Department	Year	Event Name	Registration Status
Aarav Sharma	2024PRN001	Computer Science	3rd Year	AI Tech Summit	Registered
Priya Patel	2024PRN002	Information Technology	2nd Year	Cyber Security Workshop	Pending
Rohan Deshmukh	2024PRN003	Electronics	4th Year	Robotics League 2026	Cancelled
Ananya Roy	2024PRN004	Computer Science	1st Year	CodeSprint Hackathon	Registered
Vikram Singh	2024PRN005	Mechanical Eng.	3rd Year	CAD Expo	Pending



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/* Sonal Nakhate 24070521199 */

Event Registration Table

Student event registration manager with real-time text search and status filter dropdown.

Search: Search name, PRN, event... Status Filter: Registered Try Load via fetch() Try Load via \$.getJSON()

Student Name	PRN	Department	Year	Event Name	Registration Status
Aarav Sharma	2024PRN001	Computer Science	3rd Year	AI Tech Summit	Registered
Ananya Roy	2024PRN004	Computer Science	1st Year	CodeSprint Hackathon	Registered

/* Sonal Nakhate 24070521199 */

Event Registration Table

Student event registration manager with real-time text search and status filter dropdown.

Search: Search name, PRN, event... Status Filter: Pending Try Load via fetch() Try Load via \$.getJSON()

Student Name	PRN	Department	Year	Event Name	Registration Status
Priya Patel	2024PRN002	Information Technology	2nd Year	Cyber Security Workshop	Pending
Vikram Singh	2024PRN005	Mechanical Eng.	3rd Year	CAD Expo	Pending

/* Sonal Nakhate 24070521199 */

Event Registration Table

Student event registration manager with real-time text search and status filter dropdown.

Search: Search name, PRN, event... Status Filter: Cancelled Try Load via fetch() Try Load via \$.getJSON()

Student Name	PRN	Department	Year	Event Name	Registration Status
Rohan Deshmukh	2024PRN003	Electronics	4th Year	Robotics League 2026	Cancelled



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4. Result/Conclusion: This event registration web page delivers a simple, lightweight solution for event coordinators to manage student records dynamically without manual data entry.

Key Highlights:

- Flexible Data Handling: Supports both standard Vanilla JavaScript (`fetch()`) and jQuery (`$.getJSON()`) for dynamic loading, pre-loaded with fallback sample data to eliminate initial loading errors.
- Streamlined Filtering: Features real-time text search and a dedicated status dropdown (`Registered`, `Pending`, `Cancelled`) in the top filter bar for instant record lookup.
- Clean Table Layout: Displays student details—Name, PRN, Department, Year, Event Name, and Registration Status—as readable text badges instead of cluttered form inputs.
- Robust Error & Empty States: Provides clear, contextual alerts only when external file fetches fail or when search/filter combinations return no matching records.